Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Consider the function $f(x) = x^2 - 10x + 25$. Prepare a table of values to plot the graph of $f(x)$. 1 / 1 point

Select all statements which reflect essential aspects to consider when preparing the table of values for this function.

- [x] Rewrite the expression of $f(x)$ by 'completing the square' in order to find the coordinates of the vertex

  Nice work

- [ ] Include positive and negative values of x
- [x] Include values of x symmetrical with respect to 5

  Nice work

- [x] The graph of $f(x)$ has a vertex at $(5, 0)$

  Nice work

  This function can be rewritten as $f(x) = (x-5)^2$.

- [ ] None of the above are correct

2. Write the condition that describes the interval for values of x: 1 / 1 point

$(-7, 7]$.

- ( ) $-7 \leq x \leq 7$
- (•) $-7 < x \leq 7$
- ( ) $-7 \leq x < 7$
- ( ) None of the above are correct

  Nice work

3. Let $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2 + 1$ 1 / 1 point

- [ ] The function f is a one to one function (injective function)
- [x] The function f is not a one to one function (not injective function)

  Nice work

- [ ] f is an onto (surjective function)
- [x] f is not an onto (not surjective function)

  Nice work

- [x] $\text{Domain\_f} = R$

  Nice work

- [ ] $\text{Domain\_f} = R \;\; \text{s.t} \; R \geq \{1\}$

4. Given the kinematics equations 1 / 1 point

$$s = \frac{u+v}{2}t$$

$$v = u + at$$

$$s = ut + \frac{at^2}{2}$$

$$v^2 = u^2 + 2as$$

where

s is displacement from the start

u is initial velocity

v is final velocity

a is acceleration (always constant for these equations)

t is time from the start

A billiard ball is decelerating at a constant rate. Its initial velocity is 8m/s. After 1.5 seconds it is travelling at 6m/s. Find its acceleration.

- ( ) $a = 1.5\,\text{m}\,s^{-2}$
- ( ) $a = 6\,\text{m}\,s^{-2}$
- (•) $a = -1.33\,\text{m}\,s^{-2}$
- ( ) None of the above are correct

  Nice work

  $v = u + at$

  $6 = 8 + a \times 1.5$

  $a = -\frac{2}{1.5} = -1.33\,\text{m}s^{-2}$

5. Calculate the distance c between the points (3,2) and (9,7). 1 / 1 point

- ( ) c=8.81
- ( ) c=8.71
- (•) c=7.81
- ( ) None of the above are correct

  Nice work